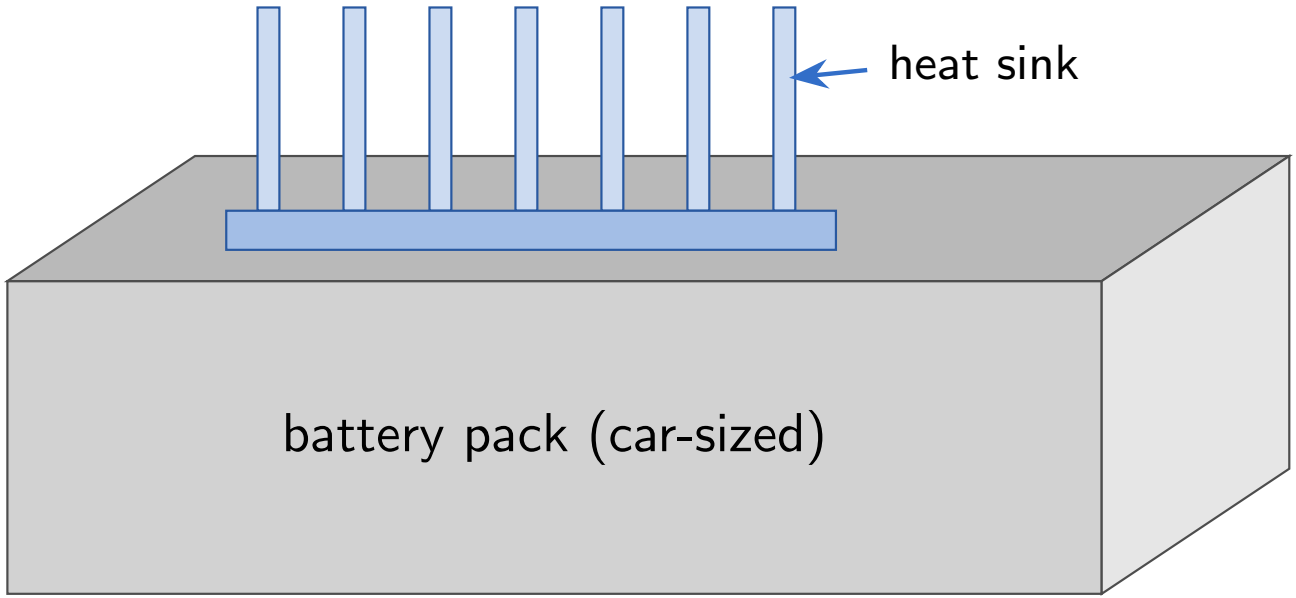
SAGE in Pictures

Faster PINN training for battery heat-sink simulation

Sayed Muhsin
Supervised by Prof. Seok-Bum Ko

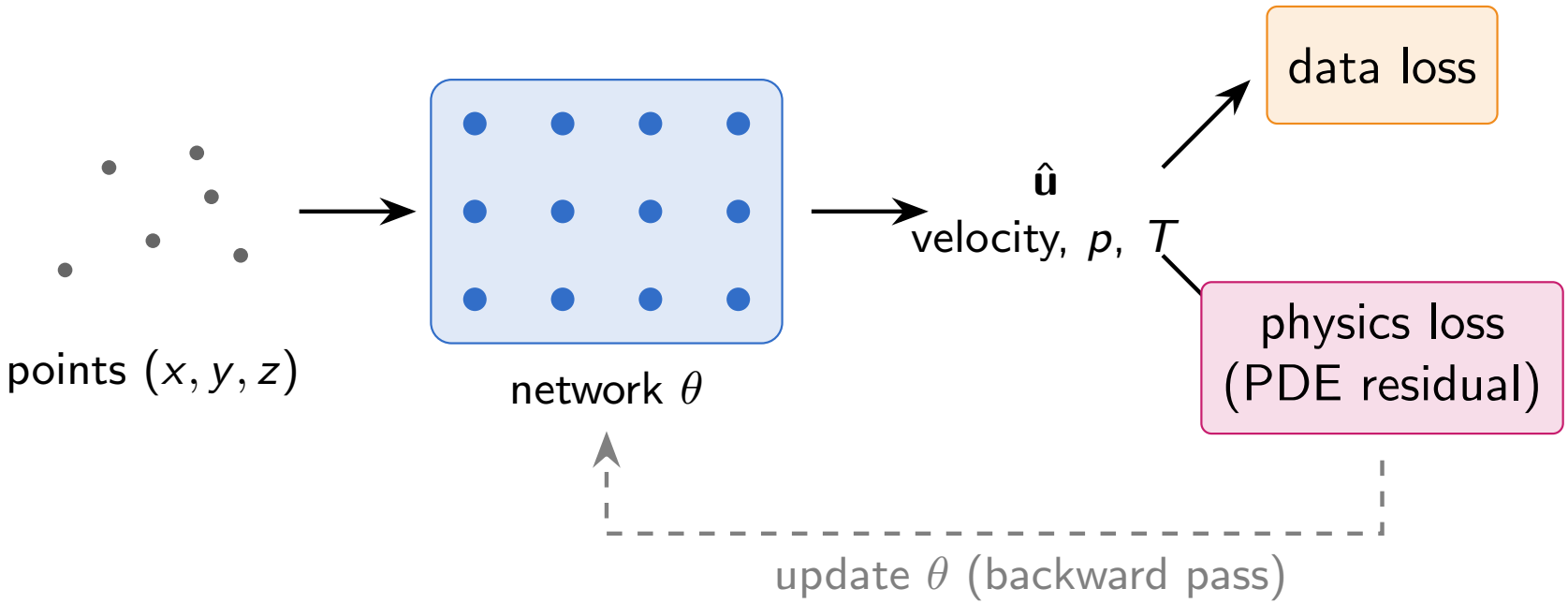
November 2026

The problem: a heat sink on a big battery

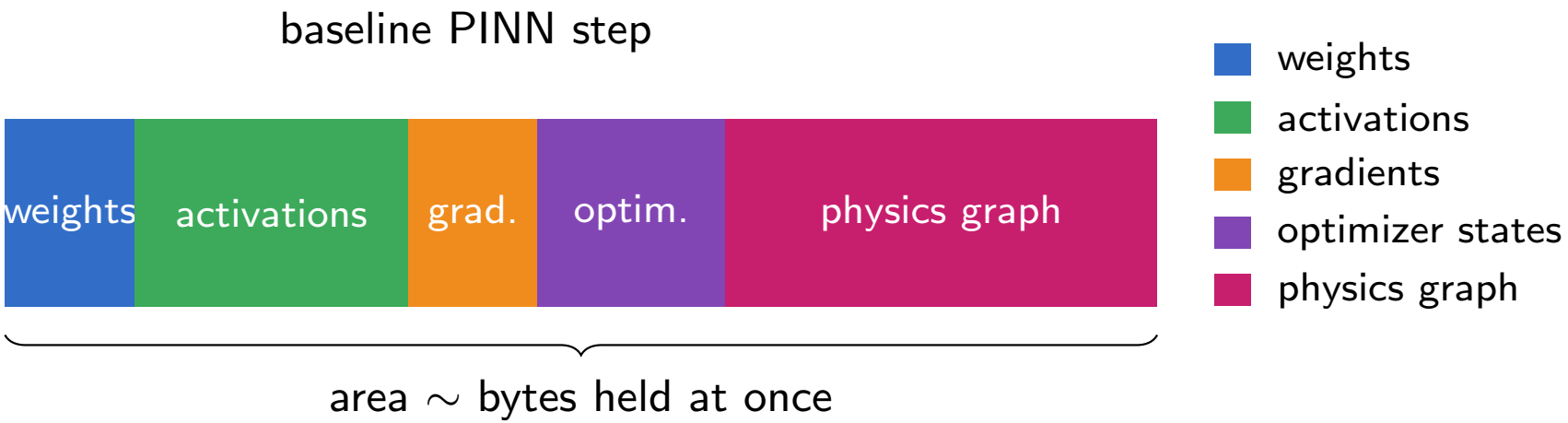


PINN predicts the physics: turbulent flow and heat (RANS k–ε)

One PINN training step

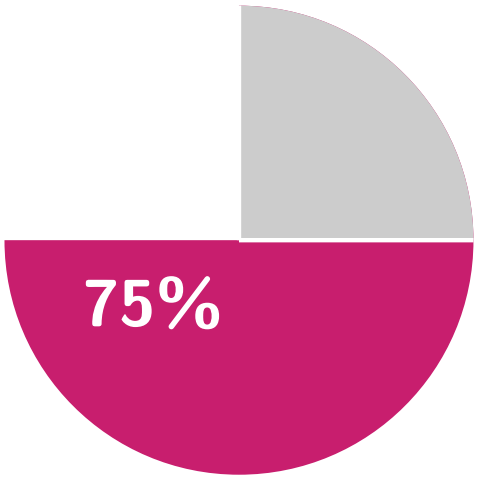


Where the memory goes in one step



Illustrative proportions, not measured.

Most of the step waits on memory

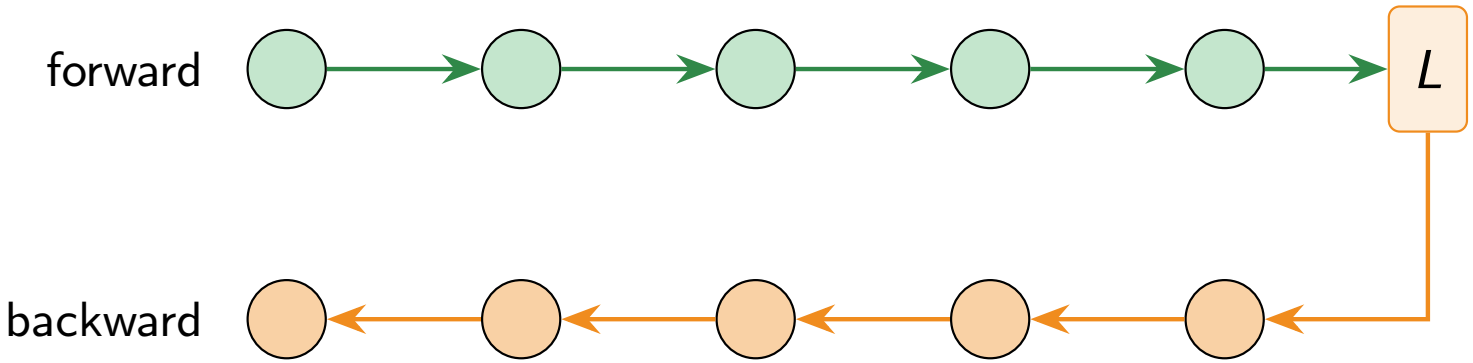


network + derivatives
at physics points

memory-bandwidth-bound

Measured, September profiling. Fix the traffic, not the maths.

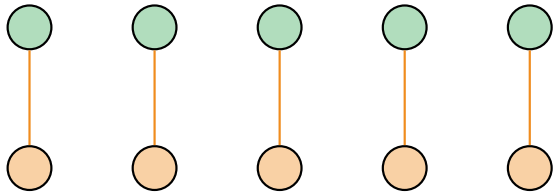
Weight gradient: one backward chain



$\partial L/\partial\theta$: one chain, $\sim$ one node per layer

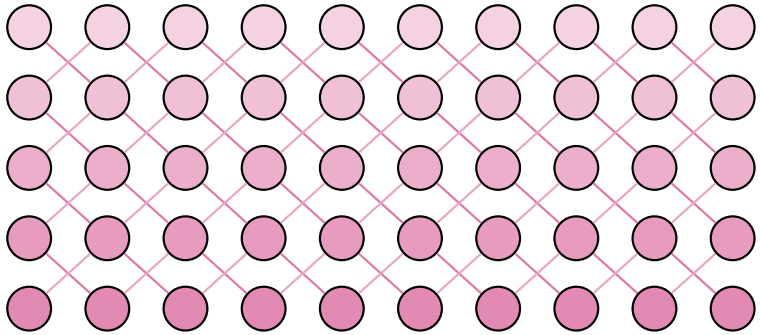
Physics gradient: derivative of a derivative

weight gradient



~10 nodes (illustrative)

physics-loss gradient



~50+ nodes, all stored (illustrative)

Needs $\partial\hat{u}/\partial x$ and $\partial^2\hat{u}/\partial x^2$, then $\partial/\partial\theta$ of those.

SAGE idea: split the gradient in two

$$\frac{\partial L}{\partial \theta} = \underbrace{\mathbf{v}}_{\partial L/\partial \hat{\mathbf{u}}} \cdot \underbrace{J}_{\partial \hat{\mathbf{u}}/\partial \theta}$$

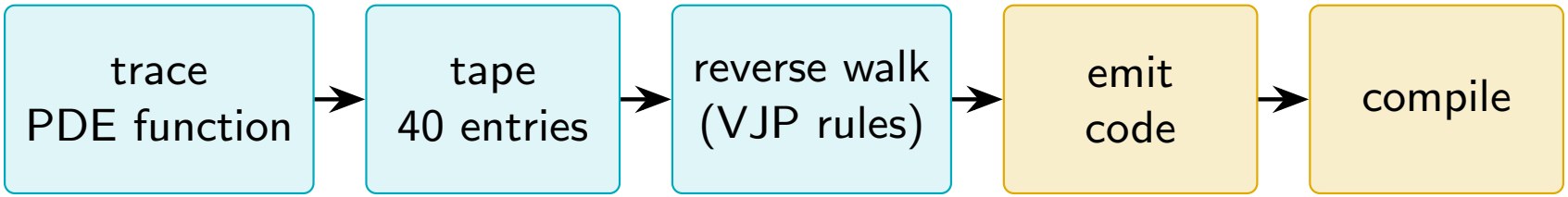
v: PDE only
fixed spectral matrices
derived symbolically, once

→

J: one ordinary backprop
no derivative-of-derivative graph

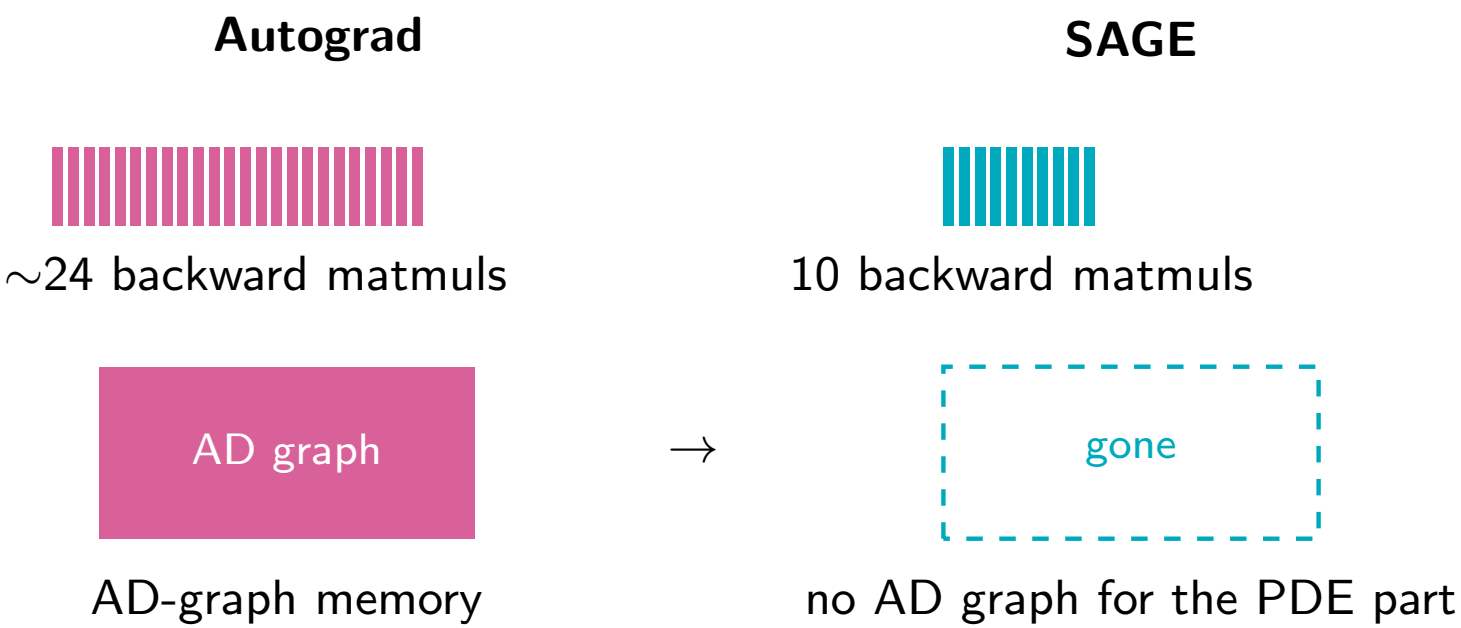
Prerequisite: DT-PINN, derivatives as fixed matrices.

How **v** is derived (once, before training)

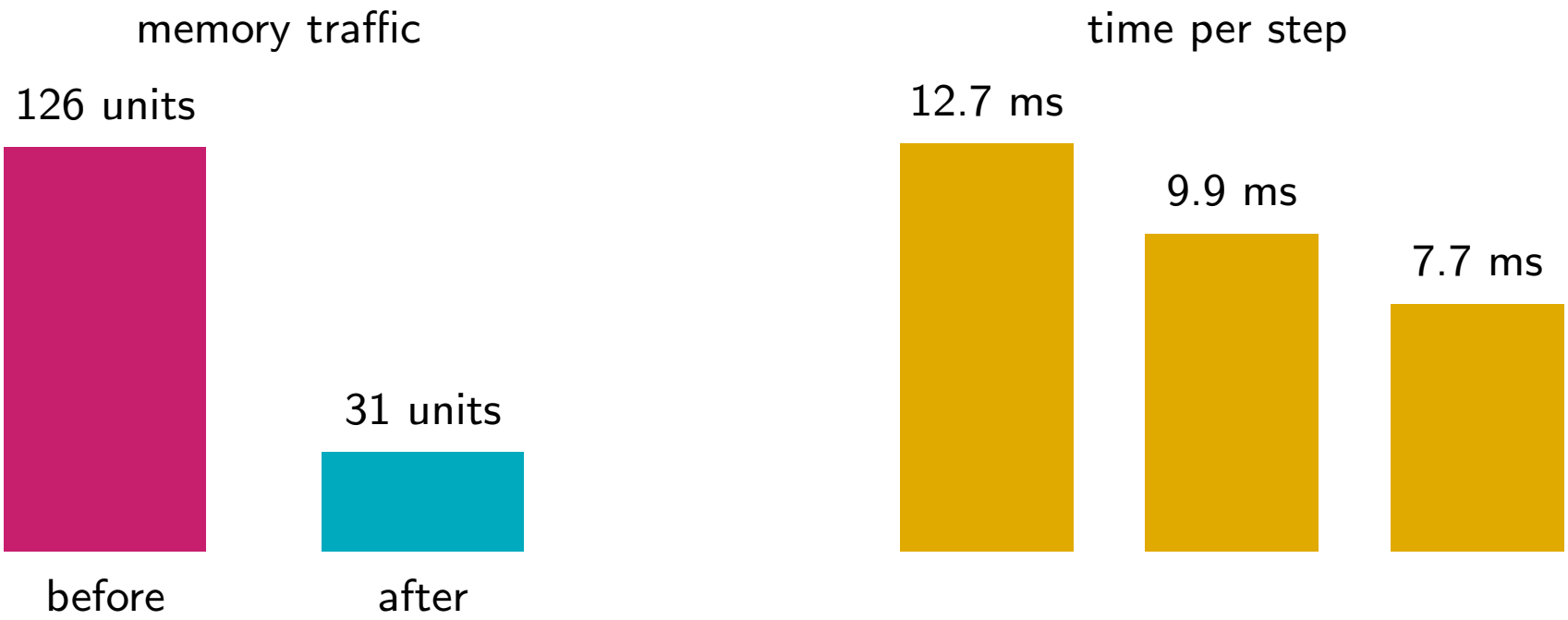


Navier-Stokes + Smagorinsky example. Structure only, no numbers.

Before and after

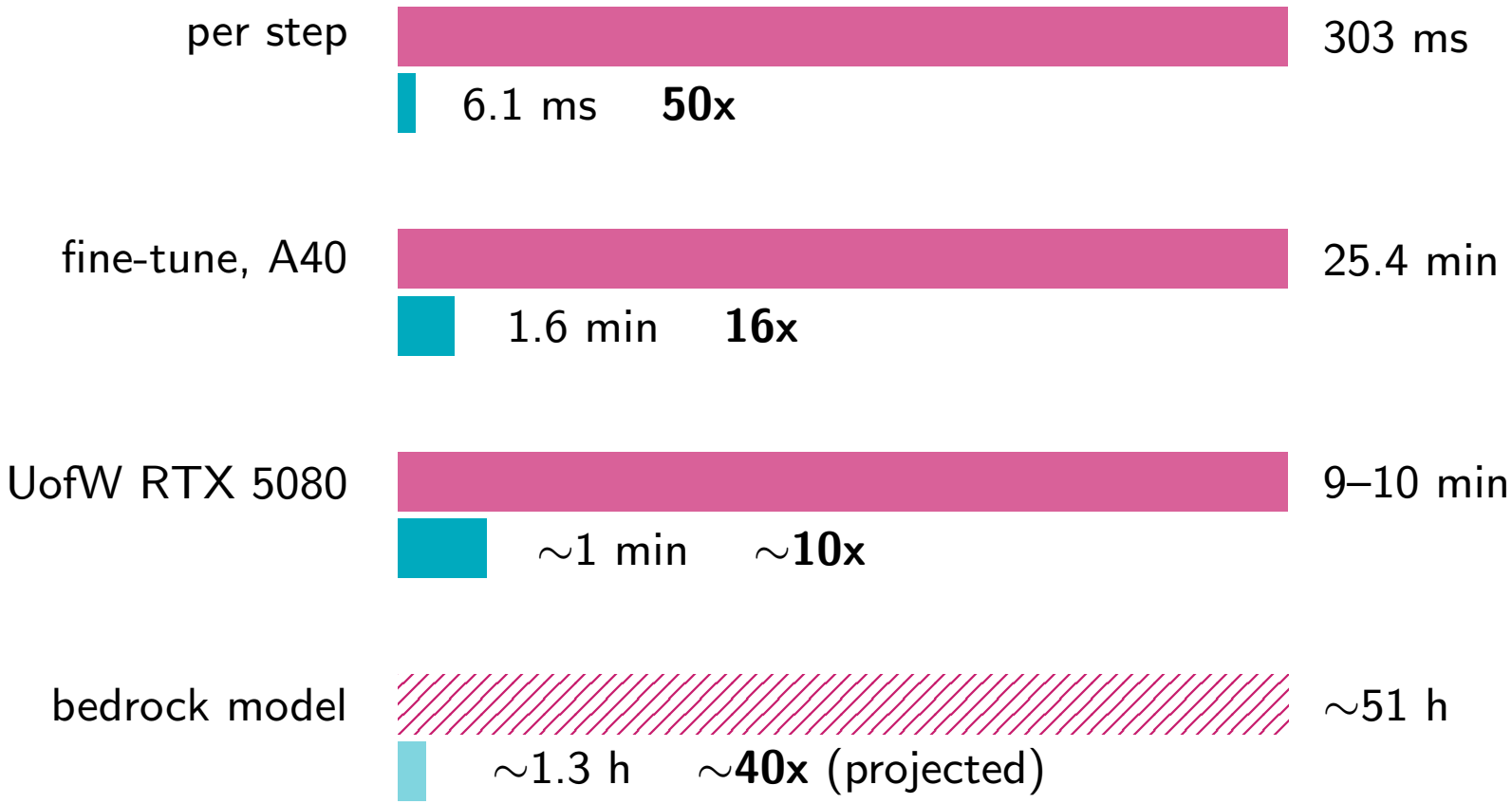


September: less memory traffic, faster step



Measured, September deck.

October: results in the UofW code



Setup time (12 min bedrock, 1 min fine-tune) is not accelerated. Bedrock figures are projected.

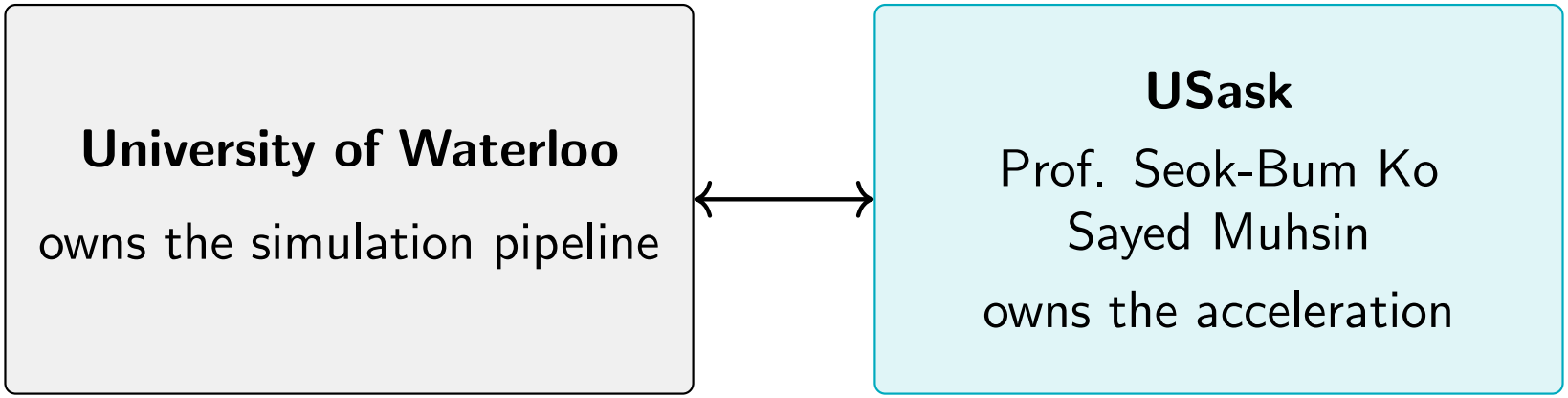
Live demo

One training step in 3D

toggle SAGE on and off

Open `demo/index.html`

Who does what



Next: support UofW's first full bedrock run; report its measured speedup.